

Yijing Wu

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EDUCATION

Northeastern University | Seattle, WA, U.S.

September 2021 – Present

Master of Science in Information Systems; GPA: 3.9/4.0

- Courses: Web Design and User Experience Engineering, Smartphones-Based Web Development

Tongji University | Shanghai, China

September 2017 – June 2021

Bachelor of Engineering in Communication Engineering (Electronic Engineering)

- Courses: Algorithms Design and Analysis, Software Technology, Principles of Database Systems, Computer Communication Networks, Brain and Cognitive Computing, Principles of Artificial Intelligence

SKILLS

- **Programming Languages:** JAVA, JavaScript/TypeScript, Python, HTML, CSS, SQL, C/C++, Swift
- **Tools/Framework:** React, React Native, Node.js, MySQL, MongoDB, Vue.js, Oracle, Docker, AWS, GCP, Git

WORK EXPERIENCE

Upduo | Berkeley, CA, U.S.

June 2022 – Present

Software Engineer Intern (React, React Native, Django, GraphQL, TypeScript, Python, Java, Objective-C)

- Worked as a full-stack engineer to build and improve Upduo's web/mobile app via React, React Native and Django
- Integrated iOS framework CallKit and Android ConnectionService via Callkeep and modified the native code with Objective-C and Java to realize a video call notification feature
- Performed local camera and microphone selection and check features through Agora to ensure video chat quality
- Collaborated with product and design team to conduct visual changes and deliver new user interface, such as animation, time windows, data visualization
- Implemented beta testing and unit tests to locate Upduo web/mobile app issues

Huawei Technologies | Shanghai, China

July 2020 – September 2020

Algorithm Engineer Intern (Python, C, OpenCV)

- Applied calibration and distortion correction algorithm to process video flows of camera sensor for self-driving vehicles
- Achieved reduction of distortion correction errors from 1.395 ± 0.136 pixels to 0.055 ± 0.192 pixels through cylindrical projection and single-point mapping
- Increased processing efficiency of DSP by developing mapping table and video processing algorithms for digital signal processors based on OpenCV

PROJECT EXPERIENCE

Rubik's Cube Helper, Northeastern University (React.js, Node.js, GCP)

November 2021 - March 2022

- Delivered a dynamic and cross-browser compatible web application to bring more personalized Rubik's Cube experience
- Implemented responsive design, accessibility development, and multiple interactions met WCAG 2.1 AA standard

Digital Image Lab, Tongji University (Python, Matlab, EEGLab)

September 2019 – June 2021

- Constructed classification models of brain during working memory maintenance based on 128-channel EEG dataset for material-specific Sternberg task through feature selection and machine learning
- Extracted features of EEG signals by empirical mode decomposition (EMD) and phase space reconstruction (PSR)
- Executed Support Vector Machine (SVM), K-nearest neighbor (KNN), Random Forest (RF), and nested cross-validation
- Accomplished improvement of classification performance from 71.45% to 91.82% under SVM and published the result in an IEEE paper as the first author

Positioning and Navigation Lab, Tongji University (Python, Unity3D, EasyAR)

April 2019 – June 2021

- Built a Barrier-Free Positioning and Navigation System for people with disabilities by performing image segmentation via PyTorch for indoor map construction as well as detecting obstacles based on the YoloV3 algorithm
- Guided a four-person research team and participated in the development of AR assisted module for indoor positioning and navigation system in China International Import Expo (CIIE) in 2019 and 2020

Automatic Collection Robot for Aqua Golf, Tongji University (Python, C++)

March 2019 – March 2020

- Led a three-person team to program and constructed a robot with Raspberry Pi to enable it to collect golf balls in water
- Performed image processing and conducted golf ball positioning and recognition through image HSB using OpenCV

ACADEMIC PUBLICATIONS

Classification of EEG signals during Working-Memory Maintenance based on Phase Space Reconstruction of Empirical Mode Decomposition, Oct 2020, ISBN:978-1-6654-2299-4